

EXPERIMENT 17:

Perform Edge detection using Sobel Matrix along X axis

AIM:- Edge detection using Sobel Matrix along X axis

PROGRAM:-

```
import cv2
import numpy as np
import os

image_path = r'C:\Users\pavan\OneDrive\OpenCV\flower.png'
if not os.path.exists(image_path):
    print("Image file not found at:", image_path)
    exit()

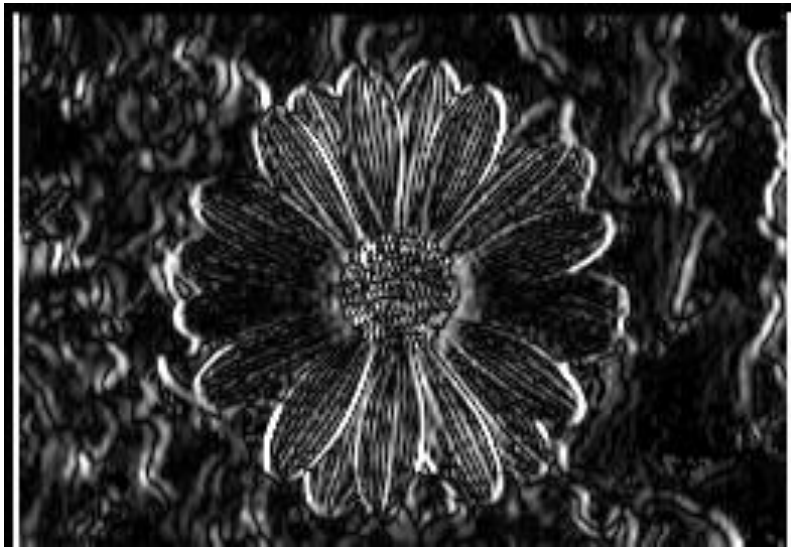
img = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)
if img is None:
    print("Failed to load image.")
    exit()

sobelx = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)
sobelx = cv2.convertScaleAbs(sobelx)
cv2.imwrite('SobelX.jpg', sobelx)
cv2.imshow('Sobel X-axis Edges', sobelx)
cv2.waitKey(0)
cv2.destroyAllWindows()
print("Sobel X-axis edge detection complete. Output saved as SobelX.jpg")
```

INPUT:-



Output:-



RESULT:-

Thus, the program is executed successfully